SPRUSON & FERGUSON

AUSTRALIA

PATENTS ACT 1990

PATENT REQUEST: STANDARD PATENT

I/We, the Applicant(s)/Nominated Person(s) specified below, request I/We be granted a patent for the invention disclosed in the accompanying standard complete specification.

[70,71] Applicant(s)/Nominated Person(s):

F.Hoffmann-La Roche AG, of 124 Grenzacherstrasse, CH-4070 Basle, SWITZERLAND

[54] Invention Title:

Preparation of Cephem- and Isooxacephem Derivatives

[72] Inventor(s):

Thomas Oberhauser

[74] Address for service in Australia:

Spruson & Ferguson, Patent Attorneys
Level 33 St Martins Tower
31 Market Street
Sydney New South Wales Australia (Code **SF**)

Details of Basic Application(s):

[31] Appl'n No(s):	[33] Country:	[32] Application Date:
EP96109239.2	CH	10 June 1996

DATED this SIXTH day of JUNE 1997

F.Hoffmann-La Roche AG

By:

Registered Patent Attorney

IRN: 380440

INSTR CODE: 55541

S076589 06 JUN97

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Ref. RAN 4410/252

Australia

Patents Act 1990

NOTICE OF ENTITLEMENT

I, Roland Borer

of 10 Stockackerstrasse, CH-4153 Reinach, Switzerland

being authorised by the Applicant(s)/Nominated Person(s) in respect of an application entitled:

Preparation of Cephem- and Isooxacephem Derivatives

state the following:-

1. The Applicant(s)/Nominated Person(s) has/have, for the following reasons, gained entitlement from the actual inventor(s):-

[X] The Applicant is the assignee of the invention from the inventor(s)

2a.* The Applicant(s)/Nominated Person(s) is/are the applicant(s) of the basic application(s) listed* on the Patent Request/* ~~in the Declaration under Article 8 of the PCT.~~

2b.* The Applicant(s)/Nominated Person(s) is/are entitled to rely on the basic application(s) listed* on the Patent Request/~~* in the Declaration under Article 8 of the PCT as follows:~~

3.* The basic application(s) listed* on the Patent Request/~~* in the Declaration under Article 8 of the PCT~~ is/are the applications first made in a Convention Country in respect of the invention.

DATED this 13th day of May 1997.

Roland Borer

* Delete if not applicable

AU9724746

(12) PATENT ABRIDGMENT (11) Document No. AU-B-24746/97
(19) AUSTRALIAN PATENT OFFICE (10) Acceptance No. 697970

(54) Title
PREPARATION OF CEPHEM- AND ISOOXACEPHEM DERIVATIVES

International Patent Classification(s)
(51)[6] **C07D 501/04** **C07D 501/24** **C07D 507/08** **C07F 009/165**

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(43) Publication Date : **18.12.97**

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(71) Applicant(s)
F.HOFFMANN-LA ROCHE AG

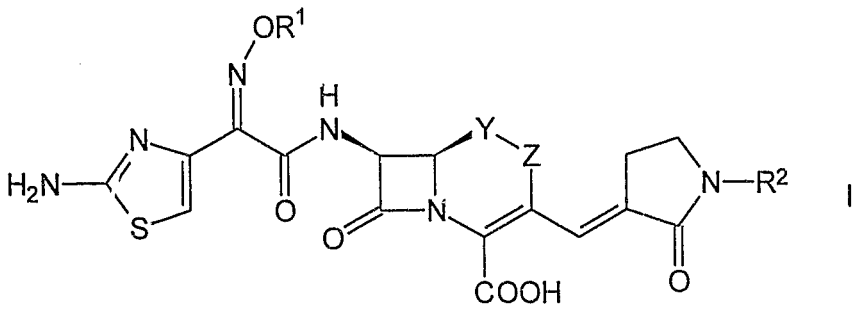
(72) Inventor(s)
THOMAS OBERHAUSER

(74) Attorney or Agent
SPRUSON & FERGUSON , GPO Box 3898, SYDNEY NSW 2001

(56) Prior Art Documents
EP 620228
WO 96/12712
DE 2942797

(57) Claim

1. Process for the preparation of compounds of formula



I

wherein

R^1 is trityl, acetyl or tetrahydropyranyl;

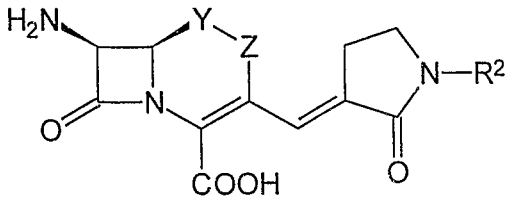
R^2 is hydrogen, hydroxy, lower alkyl, cycloalkyl, lower alkoxy, lower alkenyl, lower alkynyl, aryl, aryloxy, aryl-lower alkyl, aryl-lower alkoxy, heterocyclyl or heterocyclyl-lower alkyl; the lower alkyl, cycloalkyl, lower alkoxy, lower alkenyl, cycloalkenyl, lower alkynyl, aryl-lower alkyl, aryl, aryloxy, aryl-lower alkoxy, the heterocyclyl moieties being unsubstituted or substituted with at least one group selected from carboxy, amino, nitro, cyano, lower alkyl, lower alkoxy, hydroxy, halogen, $-CONR^{21}R^{22}$, $-N(R^{22})COOR^{23}$, $R^{22}CO-$, $R^{22}OCO-$ or $R^{22}COO-$, wherein R^{21} is hydrogen, lower alkyl, or cycloalkyl; R^{22} is hydrogen or lower alkyl; R^{23} is lower alkyl, lower alkenyl or a carboxylic acid protecting group;

.../2

Y is -S- and Z is -CH_2- or

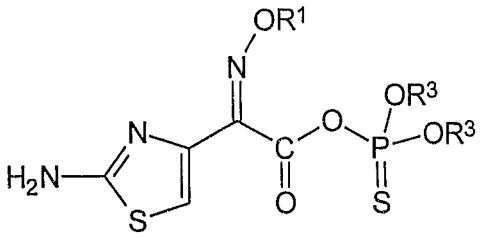
Y is -CH_2- and Z is -O-,

by acylation of a compound of formula



II

with an activated carboxylic acid derivative of formula

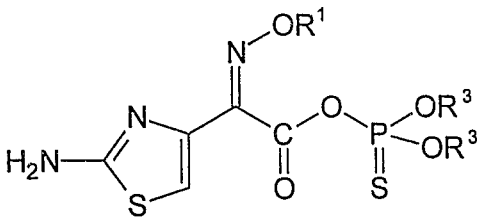


III

wherein R^3 is lower alkyl, and R^1, R^2, Y, Z have the significance given above.

5. A 7-[2-(2-aminothiazol-4-yl)-2-hydroxyimino-acetylamino]-3-(2-oxo-pyrrolidin-3-ylidenemethyl)-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid or derivative thereof, prepared by the process of any one of claims 1 to 4.

6. Compounds of formula



III

wherein R^1 is trityl, acetyl or tetrahydropyranyl and R^3 is lower alkyl.

S & F Ref: 380440

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COMPLETE SPECIFICATION

FOR A STANDARD PATENT

ORIGINAL

Name and Address of Applicant: F.Hoffmann-La Roche AG
124 Grenzacherstrasse
CH-4070 Basle
SWITZERLAND

Actual Inventor(s): Thomas Oberhauser

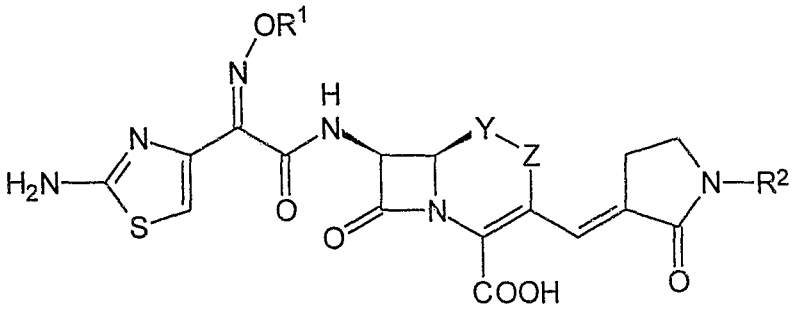
Address for Service: Spruson & Ferguson, Patent Attorneys
Level 33 St Martins Tower, 31 Market Street
Sydney, New South Wales, 2000, Australia

Invention Title: Preparation of Cephem- and Isooxacephem Derivatives

The following statement is a full description of this invention, including the best method of performing it known to me/us:-

Preparation of Cephem- and Isooxacephem Derivatives

The present invention is concerned with a process for the preparation of cephem- and isooxacephem derivatives of formula



I

wherein

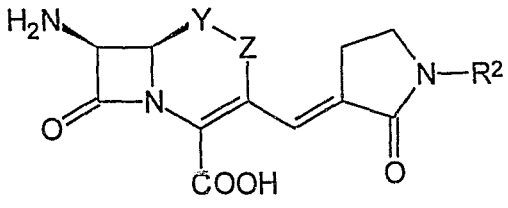
R^1 is trityl, acetyl or tetrahydropyranyl;

R^2 is hydrogen, hydroxy, lower alkyl, cycloalkyl, lower alkoxy, lower alkenyl, lower alkynyl, aryl, aryloxy, aryl-lower alkyl, aryl-lower alkoxy, heterocyclyl or heterocyclyl-lower alkyl; the lower alkyl, cycloalkyl, lower alkoxy, lower alkenyl, cycloalkenyl, lower alkynyl, aryl-lower alkyl, aryl, aryloxy, aryl-lower alkoxy, the heterocyclyl moieties being unsubstituted or substituted with at least one group selected from carboxy, amino, nitro, cyano, lower alkyl, lower alkoxy, hydroxy, halogen, $-CONR^{21}R^{22}$, $-N(R^{22})COOR^{23}$, $R^{22}CO-$, $R^{22}OCO-$ or $R^{22}COO-$, wherein R^{21} is hydrogen, lower alkyl, or cycloalkyl; R^{22} is hydrogen or lower alkyl; R^{23} is lower alkyl, lower alkenyl or a carboxylic acid protecting group;

Y is -S- and Z is $-CH_2-$ or

Y is $-CH_2-$ and Z is -O-,

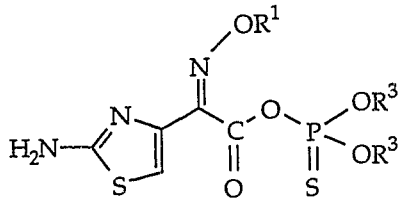
by acylation of a compound of formula



II

with an activated carboxylic acid derivative of formula

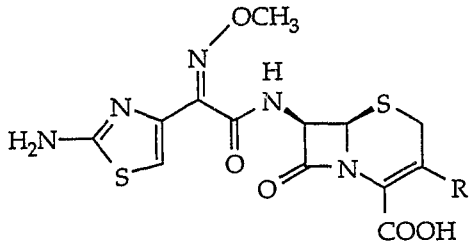
AUSTRALIAN PATENT OFFICE



III

wherein R^3 is lower alkyl, and R^1, R^2, Y, Z have the significance given above.

Cephalosporine derivatives having a methoxyimino-aminothiazole unit as depicted below



may be prepared in a convergent manner i.e. by acylation of the cephem unit with an activated aminothiazole derivative. The activating group can be a benzothiazolyl ester as described in J. Antibiotics 1990, 43, 1564-1572 or it can be a mixed anhydride of thiophosphoric acid as described in EP-A-0 620 228.

As used herein, the term "lower alkyl" refers to both straight and branched chain saturated hydrocarbon groups having 1 to 8, preferably 1 to 4 carbon atoms, for example, methyl, ethyl, n-propyl, isopropyl, tertiary butyl and the like.

By the term "cycloalkyl" is meant a 3-7 membered saturated carbocyclic ring e.g., cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl and the like.

The term "lower alkoxy" refers to an oxygen radical having an alkyl group as defined above, examples include methoxy, ethoxy, n-propyloxy and the like.

As used herein, "lower alkenyl" refers to an unsubstituted or substituted hydrocarbon chain radical having from 2 to 8 carbon atoms, preferably from 2 to 4 carbon atoms, and having at least one olefinic double bond, e.g. vinyl, allyl and the like.

As used herein, "lower alkynyl" refers to an unsubstituted or substituted hydrocarbon chain radical having from 2 to 8 carbon atoms, preferably 2 to 4 carbon atoms, and having at least one olefinic triple bond.

The term "halogen" used herein refers to all four forms, that is chlorine or chloro; bromine or bromo; iodine or iodo; and fluorine or fluoro.

By the term "aryl" is meant a radical derived from an aromatic hydrocarbon by the elimination of one atom of hydrogen and can be substituted or unsubstituted. The aromatic hydrocarbon can be mononuclear or polynuclear. Examples of aryl radicals of the mononuclear type include phenyl, tolyl, xylyl, mesityl, cumenyl, and the like. Examples of aryl radicals of the polynuclear type include naphthyl, anthryl, phenanthryl, and the like. The aryl group can have at least one substituent selected from, as for example, halogen, hydroxy, cyano, carboxy, carbamoyl, nitro, amino, aminomethyl, lower alkyl, lower alkoxy or trifluormethyl. Examples include 2-fluorophenyl, 3-nitrophenyl, 4-nitrophenyl, 4-methoxyphenyl, 4-hydroxyphenyl and the like.

By the term "aryl-lower alkyl" is meant a lower alkyl group containing an aryl group as defined above, for example benzyl.

As used herein, "aryloxy" is an oxygen radical having an aryl substituent as defined above (i.e., -O-aryl).

As used herein, "aryl-lower alkoxy" is an oxygen radical having an aryl-lower alkyl substituent. (i.e., -O-lower-alkyl-aryl).

As used herein, "heterocyclyl ring" refers to the residue of an unsaturated or saturated, unsubstituted or substituted 5-, 6-, or 7-membered heterocyclic ring containing at least one hetero atom selected from the group consisting of oxygen, nitrogen, or sulfur. Exemplary heterocyclyl groups include, but are not limited to, e.g., the following groups: pyridyl, pyridiniumyl, pyrazinyl, piperidyl, piperidino, N-oxido-pyridyl, pyrimidyl, piperazinyl, pyrrolidinyl, pyridazinyl, N-oxide-pyridazinyl, pyrazolyl, triazinyl, imidazolyl, thiazolyl, 1,2,3-thiadiazolyl, 1,2,4-thiadiazolyl, 1,3,4-thiadiazolyl, 1,2,5-thiadiazolyl, 1,2,3-oxadiazolyl, 1,2,4-oxadiazolyl, 1,3,4-oxadiazolyl, 1,2,5-

oxadiazolyl, 1,2,3-triazolyl, 1,2,4-triazolyl, 1H-tetrazolyl, 2H-tetrazolyl, thienyl, furyl, hexamethyleneiminyl, oxepanyl, 1H-azepinyl, thiophenyl, tetrahydrothiophenyl, 3H-1,2,3-oxathiazolyl, 1,2,3-oxadiazolyl, 1,2,5-oxadithiolyl, isoxazolyl, isothiazolyl, 4H-1,2,4-oxadiazinyl, 1,2,5-oxathiazinyl, 1,2,3,5-oxathiadiazinyl, 1,3,4-thiadiazepinyl, 1,2,5,6-oxatriazepinyl, oxazolidinyl, tetrahydrothienyl, and others. Substituents for the heterocyclic ring include lower-alkyl, lower- alkoxy, halogen, trifluoromethyl, trichloroethyl, amino, mercapto, hydroxy, carboxy or carbamoyl. Preferred examples of substituted heterocyclyl groups are mono-substituted and include 5-methyl-isoxazol-3-yl, N-methyl-pyridinium-2yl, 1-methyl-tetrazolyl and the like.

As used herein, "heterocyclyl-lower alkyl" refers to a lower alkyl group containing a heterocyclic group as defined above, e.g. tetrazolyl-methyl, tetrahydrofuranyl-methyl, thiophenyl-methyl or benzimidazolyl-methyl.

The heterocyclic ring can also be substituted by an optionally substituted phenyl ring such as 2,6-dichlorophenyl. Preferred is 2,6-dichlorophenyl-5-methyl-isoxazolyl.

A further substituent of the heterocyclic ring is oxo, such as in 2-oxo-oxazolidin-3-yl, 1,1-dioxo-tetrahydrothien-3-yl.

The heterocyclic ring can also be fused together with a benzene ring.

By the term "substituted phenyl" is meant phenyl mono or di-substituted.

The process according to the invention is especially suited for the preparation of cephem and isocephem derivatives respectively of formula I wherein R^1 is hydrogen, i.e. with an hydroxyimino group, which has to be protected during the acylation step. It is essential that the protecting groups are cheap, easily removeable, recycleable and that no additional purification steps are involved due to contamination of a catalyst used during the protecting and deprotecting process. Furthermore the protecting group should not interfer with the acylation step.

It has been found that the acylation process according to invention is especially suited for the acylation of cephem- and isooxacephem derivatives of formula II with an aminothiazol derivative of formula III which is activated as mixed anhydride of thiophosphoric acid and R^1 is protected by a trityl, acetyl or tetrahydropyranyl group, preferably a trityl group. The yield of this reaction as well as the purity of the product are excellent and the protecting groups are easily removed to yield hydroxyimino compounds, i.e. compounds of formula I wherein R^1 is hydrogen.

The acylation of a compound of formula II with the activated compound of formula III is preferably carried out in a polar solvent as dimethyl formamide (DMF), dichloromethane, or a mixture of DMF/i-pronanol/water in presence of a base as e.g. triethylamine, at a temperature of about -10°C to about 60°, preferably from about 0°C to about 30°C.

The compounds of formula III are new and are part of the present invention. They can be prepared as described in the Examples or in analogy thereto.

The following examples illustrate the invention in more detail and are not intented to be a limitation in any manner.

The following abbreviations were used:

mp	melting point
HPLC	high performance liquid chromatography

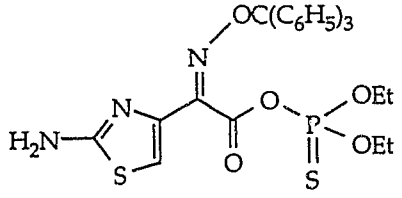
HPLC-analysis were performed as follows:

Sample preparation:	The heterogeneous reaction mixture was dissolved with a little DMSO and diluted with CH_3CN.
Instrument:	HP-1050 HPLC System.
Column:	Machery-Nagel Nucleosil 100-5 C18 AB, 250 x 4 mm.
Column temperature:	50°C.
Mobile Phase:	**A** water + 5% CH_3CN; **C** CH_3CN; **D** 0.03M potassium phosphate buffer pH 3 + 10% CH_3CN.
Gradient (t[min], **A:C:D**):	(0, 85:0:15); (8, 15:70:15); (19, 15:70:15); (19.5, 85:0:15).
Flow:	1.2 ml/min.

Detection: UV 225 nm.

Example 1

a) Preparation of (*Z*)-(2-Aminothiazol-4-yl)-trityloxyiminoacetic acid diethoxythiophosphoryl ester



To a stirred suspension of 50 g (Z)-(2-aminothiazol-4-yl)-trityloxyimino-acetic acid (116.4 mmol) and 130 mg 1,4-diazabicyclo[2.2.2]octane (DABCO) (1.164 mmol) in 500 ml dichloromethane was added under argon atmosphere 36 ml tributylamine (151 mmol). After 5 min, the red solution was cooled to 2°C. With the aid of a syringe pump was added over 30 min 24.5 ml diethyl chlorothiophosphate (151 mmol). Stirring was continued at 2°C for 1.5 h. After approximately 30 min, the activated ester (*Z*)-(2-Aminothiazol-4-yl)-trityloxyiminoacetic acid diethoxythiophosphoryl ester started to crystallize from the brown reaction mixture. The reaction was followed by HPLC. After 1 h, the starting material was consumed. To the heterogeneous reaction mixture was added dropwise over 1.5 h 750 ml water (to remove water soluble by-products) and over 40 min 500 ml *n*-hexane (to drive the precipitation of the product to completion). The suspension was stirred for 1h at 2°C and then filtered. The crystalline product was washed with 3 x 100 ml water and 3 x 100 ml *n*-hexane / dichloromethane 3:1 and dried to constant weight. Activated ester (*Z*)-(2-Aminothiazol-4-yl)-trityloxyiminoacetic acid diethoxythiophosphoryl ester was obtained as a tan solid (64.24 g, yield = 94.9%, HPLC = 97.5 area %, mp = 146 °C) and was stored under Ar at 4°C. No further purification was necessary and the product was used as isolated for the next step.

IR (KBr) 3444, 3092, 2983, 1770, 1618, 1541, 1490, 1024, 720.

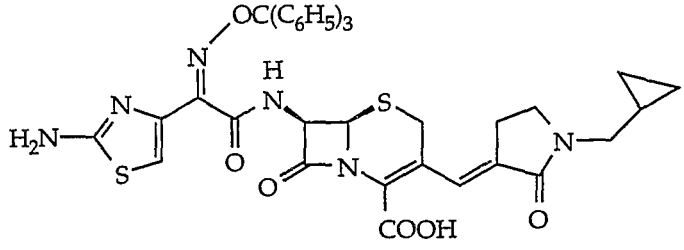
^{1}H-NMR (250 MHz, $CDCl_3$) δ 1.29 (dt,J_1=7,J_2=0.8,6H); 4.19 (dq,J_1=8.0, J_2=7.0,4H); 6.01 (s,br,2H); 6.59 (s,1H); 7.26-7.34 (m,15H).

^{31}P-NMR (100 MHz, $CDCl_3$) δ 59,05.

ISP-MS 582.4 (100, $[M+H]^+$).

MA calculated for $C_{28}H_{28}N_3O_5PS_2$ C 57.82, H 4.85, N 7.22, S 11.02, P 5.33
found C 58.09, H 4.96, N 7.21, S 10.92, P 5.35 and 0.35% water.

b) (6R, 7R)-7-[(Z)-2-(2-Aminothiazol-4-yl)-2-trityloxyimino-acetylamino)]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid triethylammonium salt



To a stirred suspension of 22.78 g (E)-(6R,7R)-7-amino-3-(1-cyclopropyl-methyl-2-oxo-pyrrolidin-3-ylidenemethyl)-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (65.2 mmol) in 160 ml dimethylformamide was added under argon 9.1 ml triethylamine (65.2 mmol) at 10°C. After 30 min, to the solution was added 48 ml 2-propanol and 3.9 ml water causing the starting material to precipitate partially. The suspension was cooled to 2°C and over 5 min was added in portions 36.68 g activated ester (Z)-(2-Aminothiazol-4-yl)-trityloxyiminoacetic acid diethoxythiophosphoryl ester (66.5 mmol). Stirring was continued at room temperature with exclusion of light for 17 h. The reaction was followed by HPLC. To the slightly turbid reaction mixture was added over 2 min 9.2 ml triethylamine (65.2 mmol, 1.0 eq) resulting in a clear, yellow solution. Reference material was added and after ca. 15 min, the reaction mixture became turbid, indicating the onset of crystallization. Stirring at room temperature was continued for 60 min and then 330 ml ethylacetate was added dropwise over 90 min. To drive crystallization to completion the suspension was cooled to 2°C and stirred for 3 h at this temperature. The suspension was filtered. The crystalline product was washed with 3 x 100 ml ice-cold ethylacetate and dried to constant weight. The cephalosporin (6R, 7R)-7-[(Z)-2-(2-Aminothiazol-4-yl)-2-trityloxyimino-acetylamino)]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid triethylammonium salt was obtained as an off-white solid (51.56 g, yield = 77%, HPLC = 100 area %) and was stored under Ar at 4°C. No further purification was necessary and the product was used as isolated for the next step.

Anal.

^{1}H-NMR (250 MHz, DMSO) δ 0.20 (m,2H); 0.46 (m,2H); 0.92 (m,1H); 3.14 (d,J=7.0,2H); 3.22-4.09 (mm,7H); 3.78, 3.82 (2d,J=16.0,2H); 5.16 (d,J=5.0,1H); 5.87 (dd,J_1=13.2,J_2=8.3,1H); 6.61 (s,1H); 7.23-7.33 (mm,16H); 9.90 (d,J=8.3,1H) + signals for NEt_3 and DMF.

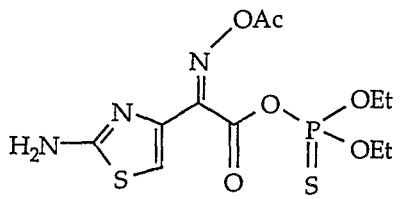
calculated for $C_{40}H_{36}N_6O_6S_2$: $C_6H_{15}N$: C_3H_7NO = 1:1:2 and 0.36 % H_2O

C 61.94, H 6.50, N 12.50, S 6.36

found C 61.49, H 6.29, N 12.17, S 6.69.

Example 2

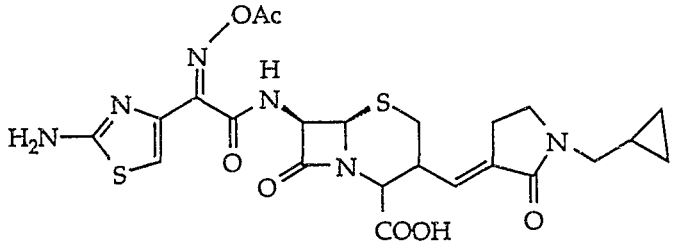
a) Preparation of (Z)-(2-Aminothiazol-4-yl)-acetoxyiminoacetic acid diethoxythiophosphoryl ester



To a stirred solution of 134.9 g (Z)-(2-aminothiazol-4-yl)-acetoxyimino-acetic acid dihydrate (508.6 mmol) and 570 mg 1,4-diazabicyclo[2.2.2]octane (DABCO) (5.09 mmol) in 1500 ml dimethylacetamide was added under argon 158 ml tributylamine (661 mmol). The yellowish solution was cooled to -20°C and over 30 min was added dropwise 104 ml diethyl chlorothiophosphate (661 mmol). Stirring was continued at -20°C for 3.5 h. The reaction was followed by HPLC. After 3 h, all starting material was consumed. The reaction mixture was allowed to warm up to 0°C and over 1.0 h was added dropwise 2200 ml water. The precipitated product was filtered, washed with water and dissolved in 800 ml dichloromethane. The aqueous layer was back-extracted with 300 ml dichloromethane. The combined organic layers were dried over 70 g sodium sulfate and concentrated under reduced pressure until the product started to crystallize. The residual solution was cooled to 2°C and 1200 ml n-hexane was added dropwise over 1 h. The resulting suspension was stirred for 1 h at 2°C and then filtered. The crystalline product was washed with n-hexane and dried to constant weight. (Z)-(2-aminothiazol-4-yl)-acetoxyiminoacetic acid diethoxythiophosphoryl ester was obtained as a white solid (166.9 g, yield = 86%, mp 128-130°C and was stored under argon at -20°C. No further purification was necessary and the product was used as isolated for the next step.

IR (KBr) 3429, 3260, 3172, 3135, 1795, 1770, 1619, 1538, 1174, 1020.
^{1}H-NMR (250 MHz, $CDCl_3$) δ 1.38 (dt,J_1=7.0,J_2=0.9,6H); 2.26 (s,3H); 4.34 (dq,J_1=8.0,J_2=7.0,4H); 6.94 (s,1H); 7.50 (s,br,2H).
^{31}P-NMR (100 MHz, $CDCl_3$) δ 59.27.
ISP-MS 404.1 (31, $[M+Na]^+$), 382.1 (100, $[M+H]^+$).
MA calculated for $C_{11}H_{16}O_6N_3PS_2$ C 34.64, H 4.23, N 11.02, S 16.81, P 8.12
found C 34.64, H 4.18, N 11.07, S 16.67, P 8.02.

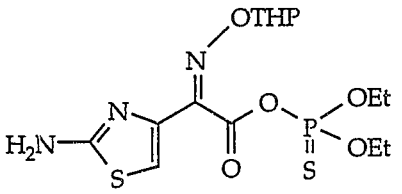
b) Preparation of (6R,7R)-7-[(Z)-2-(2-aminothiazol-4-yl)-2-acetoxyimino-acetylamino)]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidene-methyl]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid



Under an argon atmosphere to a stirred suspension of 25.6 g (E)-(6R,7R)-7-Amino-3-(1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidene-methyl)-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (73.3 mmol) in 120 ml dimethylformamide was added 20 ml triethylamine (143 mmol) at 10°C. After 15 min, the solution was cooled to 0°C and 28.5 g (Z)-(2-aminothiazol-4-yl)-acetoxyiminoacetic acid diethoxy thiophosphonyl ester (74.8 mmol) was added in portions over 5 min. Stirring was continued at 0°C with exclusion of light for 5 h. The reaction was followed by HPLC. The brown reaction mixture was poured at once into 550 ml water of 10°C. Over 30 min, 50 ml HCl 1N was added. The pH dropped from 4.6 to 3.2 and the product precipitated from the reaction mixture. Stirring was continued for 1 h at 0°C. The suspension was filtered. The product was washed with ice-cold water, re-suspended in water, stirred for 20 min at room temperature, filtered and again washed with water. (6R,7R)-7-[(Z)-2-(2-aminothiazol-4-yl)-2-acetoxyimino-acetylamino)]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid was obtained as a beige, wet solid. The product was used immediately and without drying for the next step.

Example 3

a) Preparation of (Z)-(RS)-(2-aminothiazol-4-yl)-[(tetrahydropyran-2-yloxy-imino)]-acetic acid diethoxythio-phosphoryl ester



wherein THP is tetrahydropyranyl

To a stirred suspension of 30 g (Z)-(RS)-(2-aminothiazol-4-yl)-[(tetrahydropyran-2-yloxyimino)]-acetic acid (80.5 mmol) and 90 mg 1,4-diazabicyclo[2.2.2]octane (DABCO) (0.80 mmol) in 300 ml dimethylacetamide was added under argon over 45 min 17 ml diethyl chlorothiophosphate (104.9 mmol). Stirring was continued at 0°C for 1 h. The reaction was followed by HPLC. To the slightly turbid reaction mixture was added dropwise over 50 min 450 ml water. The precipitated product was filtered, washed with water and dissolved in dichloromethane. The aqueous layer was back-extracted with dichloromethane. The combined organic layers were dried over sodium sulfate and concentrated under reduced pressure until the product started to crystallize. To the residual solution was added dropwise over 30 min n-hexane. The resulting suspension was cooled to 2°C, stirred for 1 h and then filtered. The crystalline product was washed with n-hexane and dried to constant weight. (Z)-(RS)-(2-aminothiazol-4-yl)-[(tetrahydropyran-2-yloxyimino)]-acetic acid diethoxythio-phosphoryl ester was obtained as a white solid (28.01 g, yield = 82%) and was stored under argon at -20°C. No further purificiation was necessary and the product was used as isolated for the next step.

IR (KBr) 3423, 3261, 3169, 3145, 2946, 1772, 1614, 1541, 1388, 1241, 1204, 1156, 1110, 1020, 973, 944, 908, 888, 857, 827, 727, 692.

^{1}H-NMR (250 MHz, $CDCl_3$) δ 1.37 (t,J=7.1,6H); 1.50-1.95 (m,6H); 3.65 (dm,J=11.4,1H); 3.86 (tm,J=11.4,1H); 4.33 (dq,J_1=8.0,J_2=7.0,4H); 5.47 (s,br,1H); 6.56 (s,br,2H); 6.79 (s,1H).

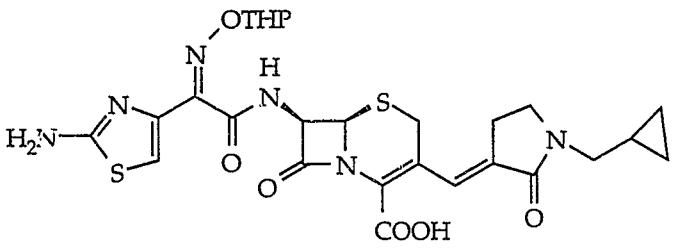
^{31}P-NMR (100 MHz, $CDCl_3$) δ 59.33.

ISP-MS 446.4 (19, $[M+Na]^+$), 424.5 (26, $[M+H]^+$), 340.2 (100).

MA calculated for $C_{14}H_{22}N_3O_6PS_2$ C 39.71, H 5.24, N 9.92, S 15.14, P 7.31

found C 39.87, H 5.20, N 10.08, S 14.99, P 7.53.

b) (6R,7R)-7-[(Z)-2-(2-Aminothiazol-4-yl)-2-[(R,S)-tetrahydropyran-2-yloxy-imino-acetylamino)]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid



Under argon atmosphere to a stirred suspension of 20 g (E)-(6R,7R)-7-Amino-3-(1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl)-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (57.2 mmol) in 140 ml dimethylformamide was added 16 ml triethylamine (114.8 mmol) at 10°C. After 10 min, to the solution was cooled to 0°C and 24.72 g (Z)-(RS)-(2-amino-thiazol-4-yl)-[(tetrahydropyran-2-yloxyimino)]-acetic acid diethoxythio-phosphoryl ester (58.4 mmol) was added in portions over 1 min. Stirring was continued at 10°C with the exclusion of light for 6 h. The reaction was followed by HPLC. The reaction mixture was poured at once into a 10°C mixture of 220 ml water and 50 ml acetone. Over 30 min, 55 ml HCl 1N was added. The pH dropped from 9.6 to 3.2 and the product precipitated from the reaction mixture. Stirring was continued for 30 min at 0°C. The suspension was filtered. The product was washed with ice-cold water and dried to constant weight. (6R,7R)-7-[(Z)-2-(2-Aminothiazol-4-yl)-2-[(R,S)-tetrahydro-pyran-2-yloxyimino-acetylamino)]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid was obtained as an off-white solid (27.7 g). The product was used as isolated for the next step.

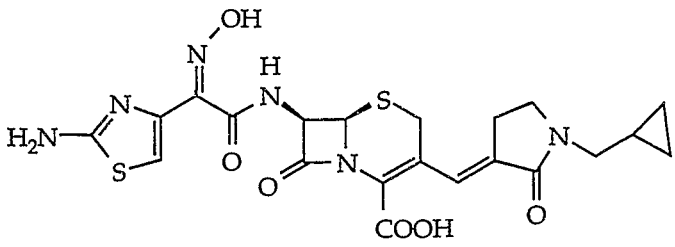
^{1}H-NMR (250 MHz, DMSO) δ 0.21 (m,2H); 0.46 (m,2H); 0.93 (m,1H); 1.40-1.90 (m,6H); 2.90-3.10 (m,2H); 3.16 (d,J=7.1,2H); 3.48 (m,2H); 3.50 (m,1H); 3.85 (m,1H); 3.90 (s,2H); 5.21 (d,J=5.0,1H); 5.26 (s,br,1H); 5.90 (dd,J_1=8.2,J_2=5.0, 1H); 6.75 (s,1H); 7.23 (s,br,3H); 9.69 (d,J=8.2,1H); 13.95 (s,br,1H).

Example 4

Cleavage of the Protective Groups:

Preparation of (6R,7R)-7-[(Z)-2-(2-aminothiazol-4-yl)-2-hydroxyimino-acetylamino]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-

oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid



a) By cleavage of the trityl group

To a stirred solution of 30 g (6R, 7R)-7-[(Z)-2-(2-Aminothiazol-4-yl)-2-trityloxyimino-acetylamino)]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid triethylammonium salt (29.2 mmol) in 60 ml dichloromethane was added over 15 min 7.5 ml triethylsilane (45.9 mmol) and over 90 min 23.9 ml trifluoroacetic acid (306 mmol) at 2°C. Stirring was continued at 10°C for 2 h. The reaction was followed by HPLC. To the reaction mixture was added over 90 min 300 ml diethylether, causing the product to precipitate. Stirring was continued for 1 h at room temperature. The suspension was filtered. The product was washed with 2 x 60 ml diethylether, again suspended in 100 ml diethylether, stirred for 15 min, filtered, washed with 2 x 40 ml diethylether and dried to constant weight. The trifluoroacetate of (6R,7R)-7-[(Z)-2-(2-aminothiazol-4-yl)-2-hydroxyimino-acetylamino]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid was obtained as an off-white solid (19.92 g, 99 %, HPLC = 100 area %) and suspended in 400 ml water. Over 15 min 20 ml NaOH 1N (20 mmol) were added at 2°C. The pH rose from 1.51 to 3.30. The suspension was stirred at 2°C for 10 min and then filtered. For the filtration a mild vacuum of about 400 mbar was applied. The product was washed with 2 x 50 ml water, suspended in 250 ml water, stirred for 15 min at 2°C, filtered, washed with 2 x 50 ml water and re-suspended in 400 ml water. Over 40 min 30 ml NaOH 1N was added at 2°C. The pH rose from 2.38 to 5.6 and most of the product dissolved. The turbid solution was filtered and two membrane filters of 0.45 µm and 0.22 µm. To the resulting, clear solution was added over 20 min 26 ml HCl 1N (26 mmol) at 2°C. The pH dropped from 5.42 to 3.30 and the product precipitated. The suspension was stirred for 60 min at 2°C, filtered and washed with 100 ml water. The product was dried (15 mbar, 24 h, 35°C) to constant weight. (6R,7R)-7-[(Z)-2-(2-Aminothiazol-4-yl)-2-hydroxyimino-acetylamino]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid

was isolated as an off white solid (12.1 g, yield 81%, HPLC 94 area %). ^{1}H-NMR (250 MHz, DMSO) δ 0.21 (m,2H); 0.46 (m,2H); 0.93 (m,1H); 2.90 (m,1H); 3.10 (m,1H); 3.15 (d,J=7.0,2H); 3.48 (t,J=6.0,2H); 3.88 (s,2H); 5.18 (d,J=4.9,1H); 5.82 (dd,J_1=8.7,J_2=4.9,1H); 6.66 (s,1H); 7.14 (s,br,2H); 7.22 (s,1H); 9.51 (d,J=8.7,1H); 11.33 (s,br,1H).

Anal.

calculated for $C_{21}H_{22}N_6O_6S_2$:

C 48.64, H 4.28, N 16.21, S 12.36

found C 47.88, H 4.36, N 15.85, S 12.17

and 2.47 % H_2O.

b) By cleavage of the acetyl group

To a stirred suspension of (6R,7R)-7-[(Z)-2-(2-aminothiazol-4-yl)-2-acetoxyimino-acetylamino)]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (used in wet form, assumed ~73.3 mmol) in 300 ml methanol was added under an argon atmosphere over 10 min 30 ml HCl conc. (304 mmol) at 2°C. After 5 h stirring at 2°C, another 10 ml HCl conc. (101 mmol) were added to the suspension. The reaction mixture was allowed to warm up to room temperature over night. The reaction was followed by HPLC. After 21 h total reaction time, all starting material was consumed and a brown solution had resulted. The reaction mixture was poured at once into 800 ml ice cold water. To the resulting suspension was added over 60 min 500 ml NaOH 1N. The pH rose from 0.6 to 3.3. Stirring at 2°C was continued for 15 min. The suspension was filtered. The product was washed with water and dried to constant weight. (6R,7R)-7-[(Z)-2-(2-Aminothiazol-4-yl)-2-hydroxyimino-acetyl-amino]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid was obtained as a yellowish solid (29.8 g, yield 78%, HPLC 90 area %).

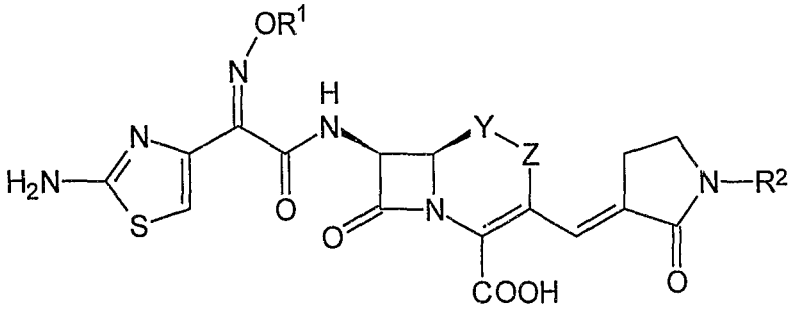
c) By cleavage of the tetrahydropyranyl (THP) group

To a stirred suspension of 20 g (6R,7R)-7-[(Z)-2-(2-aminothiazol-4-yl)-2-[(R,S)-tetrahydropyran-2-yloxyimino-acetylamino)]-3-[(E)-1-cyclo-propylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid (33.3 mmol) in 150 ml methanol was added over 10 min 15 ml HCl conc. (180 mmol) at room tempera-ture. The yellow solution was stirred at 45°C for 4.5 h. The reaction was followed by HPLC. After 4 h all starting material was consumed. The

reaction mixture was allowed to cool to room temperature and poured at once into 500 ml water. To the solution was added over 40 min 170 ml NaOH 1N. The pH rose from 0.43 to 3.1. The resulting suspension was cooled to 2°C, stirred for 1 h and filtered. The product was washed with ice cold water and dried to constant weight. (6R,7R)-7-[(Z)-2-(2-amino-thiazol-4-yl)-2-hydroxyimino-acetylamino]-3-[(E)-1-cyclopropylmethyl-2-oxo-pyrrolidin-3-ylidenemethyl]-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid was obtained as a yellowish solid (12.8 g, yield 74%, HPLC 85 area %).

The claims defining the invention are as follows:

1. Process for the preparation of compounds of formula



I

wherein

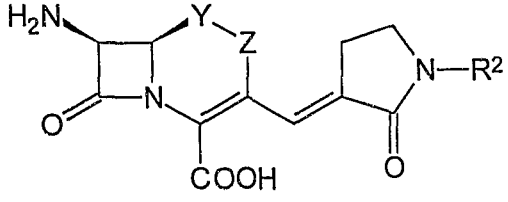
R^1 is trityl, acetyl or tetrahydropyranyl;

R^2 is hydrogen, hydroxy, lower alkyl, cycloalkyl, lower alkoxy, lower alkenyl, lower alkynyl, aryl, aryloxy, aryl-lower alkyl, aryl-lower alkoxy, heterocyclyl or heterocyclyl-lower alkyl; the lower alkyl, cycloalkyl, lower alkoxy, lower alkenyl, cycloalkenyl, lower alkynyl, aryl-lower alkyl, aryl, aryloxy, aryl-lower alkoxy, the heterocyclyl moieties being unsubstituted or substituted with at least one group selected from carboxy, amino, nitro, cyano, lower alkyl, lower alkoxy, hydroxy, halogen, $-CONR^{21}R^{22}$, $-N(R^{22})COOR^{23}$, $R^{22}CO-$, $R^{22}OCO-$ or $R^{22}COO-$, wherein R^{21} is hydrogen, lower alkyl, or cycloalkyl; R^{22} is hydrogen or lower alkyl; R^{23} is lower alkyl, lower alkenyl or a carboxylic acid protecting group;

Y is -S- and Z is $-CH_2-$ or

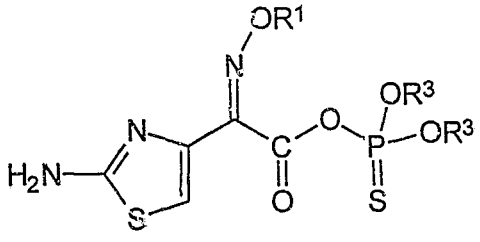
Y is $-CH_2-$ and Z is -O-,

by acylation of a compound of formula



II

with an activated carboxylic acid derivative of formula



III

wherein R^3 is lower alkyl, and R^1, R^2, Y, Z have the significance given above.

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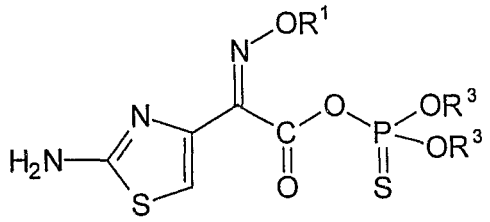
2. Process according to claim 1 for the preparation of compounds of formula I wherein Y is sulfur and Z is $-CH_2-$.

3. Process according to any one of claims 1 or 2 for the preparation of compounds of formula I wherein R^1 is acetyl or tetrahydropyranyl.

4. A process for the preparation of a 7-[2-(2-aminothiazol-4-yl)-2-hydroxyimino-acetylamino]-3-(2-oxo-pyrrolidin-3-ylidenemethyl)-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid or derivative thereof, substantially as hereinbefore described with reference to any one of the examples.

5. A 7-[2-(2-aminothiazol-4-yl)-2-hydroxyimino-acetylamino]-3-(2-oxo-pyrrolidin-3-ylidenemethyl)-8-oxo-5-thia-1-aza-bicyclo[4.2.0]oct-2-ene-2-carboxylic acid or derivative thereof, prepared by the process of any one of claims 1 to 4.

6. Compounds of formula



III

wherein R^1 is trityl, acetyl or tetrahydropyranyl and R^3 is lower alkyl.

7. A (2-aminothiazol-4-yl)-(substituted)-oxyiminoacetic acid dialkoxythiophosphoryl ester, substantially as hereinbefore described with reference to any one of the examples.

Dated 3 September 1998

F. HOFFMANN-LA ROCHE AG

Patent Attorneys for the Applicant/Nominated Person

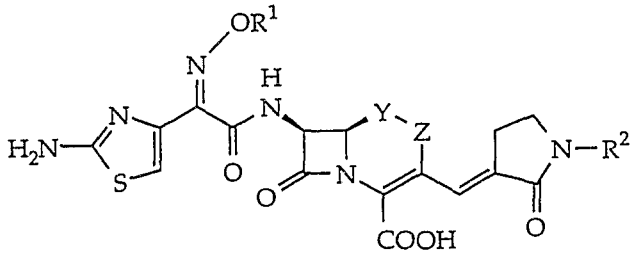
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Preparation of Cephem- and Isooxacephem

Abstract

The invention is concerned with a new process for the preparation of compounds of formula



wherein

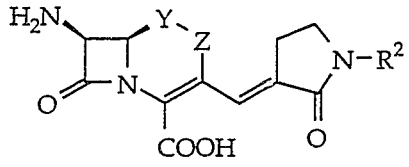
R^1 is trityl, acetyl, tetrahydropyranyl or cyclopentyl;

R^2 is hydrogen, hydroxy, lower alkyl, cycloalkyl, lower alkoxy, lower alkenyl, lower alkynyl, aryl, aryloxy, aryl-lower alkyl, aryl-lower alkoxy or heterocyclyl or heterocyclyl-lower alkyl; the lower alkyl, cycloalkyl, lower alkoxy, lower alkenyl, cycloalkenyl, lower alkynyl, aryl-lower alkyl, aryl, aryloxy, aryl-lower alkoxy, the heterocyclyl moieties being unsubstituted or substituted with at least one group selected from carboxy, amino, nitro, cyano, lower alkyl, lower alkoxy, hydroxy, halogen, $-CONR^{21}R^{22}$, $-N(R^{22})COOR^{23}$, $R^{22}CO-$, $R^{22}OCO-$ or $R^{22}COO-$, wherein R^{21} is hydrogen, lower alkyl, or cycloalkyl; R^{22} is hydrogen or lower alkyl; R^{23} is lower alkyl, lower alkenyl or a carboxylic acid protecting group;

Y is -S- and Z is $-CH_2-$ or

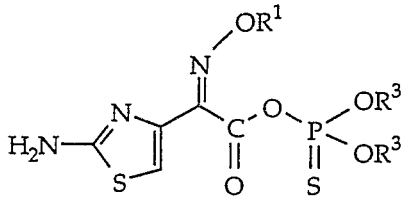
Y is $-CH_2-$ and Z is -O-,

by acylation of a compound of formula



II

with an activated carboxylic acid derivative of formula



III

wherein R^3 is lower alkyl, and R^1, R^2, X, Y, Z have the significance given above;

and it is further concerned with compounds of formula III.